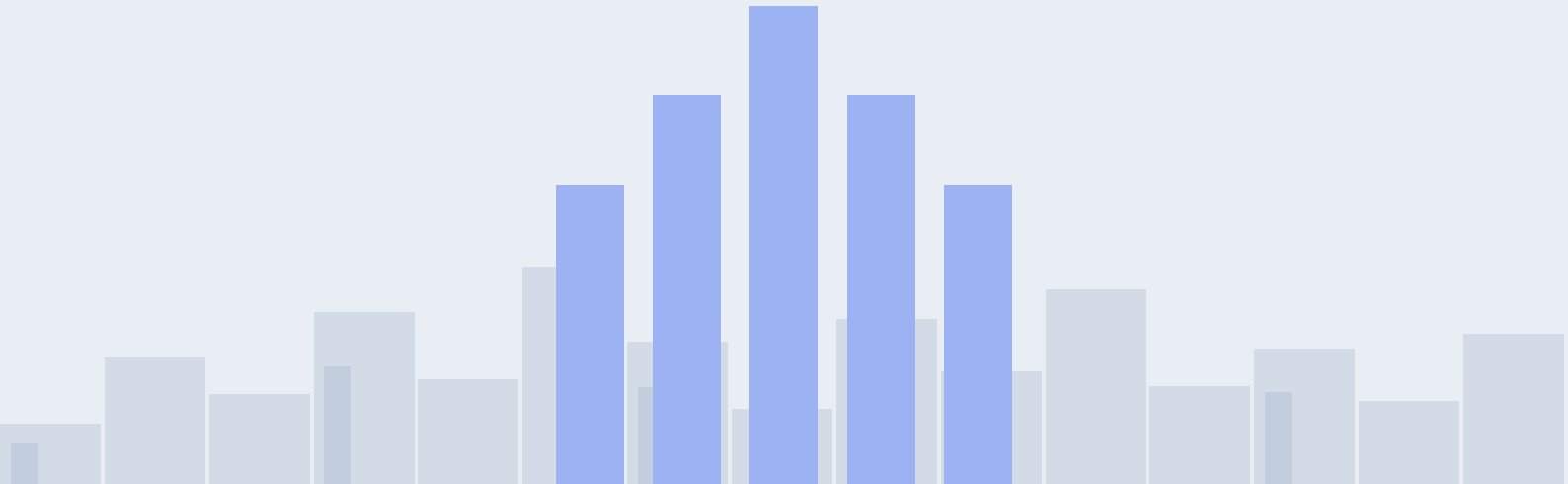
AGENT-BASED FINANCIAL MARKET SIMULATION

Market Ecology Simulator

A handbook for working, learning and understanding: what the project can do, how to operate it, which questions it answers and where it is demonstrably wrong.

324 tests · 7 agent types · 5 sectors · 7 guided experiments



1 · What the project is

An agent-based simulation of an equity market. It does not forecast prices. It asks **where market behaviour comes from**.

Around 600 agents each follow a simple rule. None knows the true value of an asset, none knows what the others will do next, and nowhere in the code is there an equation for the price. Every trade is one order meeting another. Bubbles, crashes, volatility clusters and forced-selling cascades emerge from this interaction, or they are not there.

> **The actual question**
> Real return series share a handful of statistical properties that hold across asset classes and centuries: fat tails, clustered volatility, almost no autocorrelation. None of them is built into this model. **Does a market made only of simple rules reproduce them anyway?** The criteria were fixed before the first run. That is exactly what makes the answer hold up.

What is unusual about it

- **The order book is real.** A continuous double auction with price-time priority, partial fills, cancellation and expiry. A model with `price *= (1 + k · demand)` cannot explain a liquidity hole, because it has no liquidity that could run out.
- **News never moves the price.** It moves fair value and sentiment. Whether the price follows is up to the agents, and they read the same headline differently. The ETF does not read it at all.
- **The validation is code, not prose.** Six criteria as executable checks, measured across several seeds. A result that holds for only one random start does not count.
- **Failures are documented.** Four rounds of diagnosis, several of the project's own hypotheses refuted, two criteria still open. All of it is in the README and in this handbook.

2 · Where it stands

Measured over 5 random starts × 5 instruments = 25 measurements, each 90,000 steps (about 370 trading days).

Criterion	Requirement	Measured	Measurements	
Volatility clustering	mean ACF of |returns| > 0.05	`0.217 ± 0.109`	25/25	**met**
Non-normality	Jarque-Bera p < 0.05	`0.0002`	25/25	**met**
Fat tails	excess kurtosis > 1	`5.39 ± 5.93`	24/25	**met**
Tail index	Hill estimator between 2 and 6	`4.38 ± 1.37`	22/25	**met**
No autocorrelation	|ACF(1)| < 0.1	`-0.223 ± 0.156`	8/25	**open**
Volume-volatility	correlation > 0.1	`-0.072 ± 0.129`	3/25	**open**

In addition: annualised volatility 1.31, maximum drawdown -0.62, **sector correlation 0.398** (realistic: 0.4–0.6), 786 margin calls, trend-follower share varying with a standard deviation of 0.39.

> **Why the two open points are open**
> One trade-off runs through the whole model: **the anchor that keeps a price honest is the same thing that makes its returns predictable.** Tracking fair value more closely means more reversion to the mean. Forced selling adds reversion and volatility. Every mechanism that tightened the price's grip on its fundamental value made the autocorrelation worse, every time in the same direction.

3 · The first ten minutes

```
./start.sh

# -> http://127.0.0.1:8000 the town
# -> http://127.0.0.1:8000/sectors the analysis page
```

The first start creates the virtual environment and installs the dependencies. The server answers within a few seconds and then fast-forwards the market to 110 trading days in the background. The validation panel needs 100 days before its figures mean anything, and the first few dozen days are the run settling down, not the model at work.

What to look at, in which order

Minute	Do	Watch for
1–2	Let the town on `/` run	Five blue towers in the middle are the five exchanges. Tower height = book depth. Road width = order flow. One walking figure = ten agents with an order, carrying a green (buy) or red (sell) crate.
3–4	Click a market in the market bar (TECH, ENRG …) or its tower	Price, ticker and book line follow the selection, and the selected tower turns brighter. The simulation is not affected.
5–6	Switch to `/sectors`	Correlation heatmap: configured nowhere. It emerges from the common factor in the fundamental values and from portfolios spanning several sectors.
7–8	Push the control “Trend-following strength” up	Prices detach from fair value, kurtosis in the validation panel rises. Takes effect immediately, no restart.
9–10	Set 16× and let it run	The validation panel moves towards the calibrated values. Before that it shows a shorter sample and therefore different figures.

The most important panel: “What is happening”

It sits in the town's left-hand panel and at the top of the analysis page, and it answers a fair question. A visualisation shows a **state**. Walking figures and growing blocks do not say what is happening right now and why it is worth noticing. Without this panel you would already have to know the model to read the town.

```
The market is stretched [mixed regime] [vol normal] [leverage 1.02x]

. TECH is trading +35% above fair value. Nobody decided that -
it is the result of more buying pressure than supply.
. The funds are at 1.02x leverage and have 97.3% headroom before a
margin call.

What to watch now
. The Issuer is selling new shares at this valuation. That is the
counterforce - watch whether the gap stops widening.
```

Two rules keep it from being decoration. Nothing is pre-written: a calm market is described as calm. And “high” or “unusual” is a **percentile against the run's own history**, because a volatility of 0.8 means nothing without knowing what this market normally does. Events carry their mechanism with them: not “margin call” but “A leveraged fund has fallen below its threshold and had to sell. It did not decide to.”

> **A note that saves confusion**
> The live panel shows **different figures from this handbook**. It computes on the days it has; the tables here use 370 days across five seeds. Single runs scatter considerably. The same defaults gave 5.0 of 6 on one seed over 60,000 steps, with an autocorrelation of -0.03. That is exactly why the criteria are defined across several seeds and `scripts/calibrate.py` is the figure that counts. The panel shows *which control moves what*.

4 · The seven market participants

Each exists because the model was visibly wrong without it. The reasons given are what the calibration actually showed.

Agent	Rule	Why it is in the model
Value investor	buys below its own estimate of value, sells above	the only force pulling the price towards value
Trend follower	extrapolates the recent move, ignores valuation	the feedback loop that produces bubbles
Noise trader	trades for reasons the model does not represent	background liquidity
ETF / savings plan	receives money on a schedule, buys at any price	price-insensitive demand
Leveraged fund	borrows to amplify a blended view	forced selling, the cascade mechanism
Market maker	quotes both sides, leans against its inventory	without it the book held ~600 shares against orders of several thousand: every order cleared every level and volatility was 129% a year. An artefact of the matching engine, not of behaviour
Issuer	sells new shares when expensive, buys back when cheap	without it there is no supply: value investors sell out, market makers reach their inventory limit, and then the price rises to twice fair value and stays there

Value investors and trend followers are the **same** agent type. They switch between the two rules, and that switching is the source of the interesting behaviour. It is driven by relative success (Lux & Marchesi 1999) and by herding (Kirman 1993). Herding makes the population **bistable**: it locks into a trend-following regime (volatile, bubble-prone) or a value regime (calm, mean-reverting) and stays there for a while. That persistence is what clusters volatility, and none of it is written into the price process.

5 · The guided experiments

They live in the application, not in this booklet. On `/sectors` each one sets up the market, says what to watch for, and afterwards checks by **measurement** whether it worked.

> **Why the check matters more than the instructions**
> A guided tour that always says “see, exactly as expected” teaches nothing. It would say the same if the model were broken. These report failure too, and several do so regularly. One is even designed for it: “Is this market predictable?” checks the open autocorrelation criterion and answers **not this time**, with the note that this is the model's documented flaw and not an operating error.

Experiment	Question	What it shows
Bubble	Does a bubble form on its own?	Trend following is a feedback loop. Nobody recommends anything, nobody coordinates. The bubble follows from the rules.
Cascade	Does a crash need bad news?	No. One trigger is enough. A fund below its margin threshold sells because it must, and pushes the next one below its own.
ETF	Do ETFs make the market less stable?	A savings plan does not check what anything is worth. With enough of them the price loses its anchor, not because anyone is wrong but because nobody is checking.
Correlation	Where does the correlation between sectors come from?	From a common market factor and from shared balance sheets: a fund in distress sells all its positions, not just the troubled one.
Rate turn	How does a rate hike work?	Through three channels, at different speeds. The Central Bank overshoots because the effect arrives with a delay.
Regimes	Which strategy wins in the long run?	None for good. Whichever wins attracts imitators and so makes itself unprofitable.
Forecast	Is this market predictable?	Unfortunately yes. This experiment shows the flaw on purpose instead of hiding it.

A start that pays off

- On `/sectors`, start the experiment **“Does a crash need bad news?”**.
- Switch to the town (link top right) and watch the Leverage Tower. When the cascade runs, its red beacon blinks fast, a red ring pulses on the ground and a MARGIN CALL tag appears.
- Back on the analysis page: the check says whether it worked this time, and with which figure.

6 · What the project can do

Watch it live

- **The town** (`/`) is a bright 3D diorama (WebGL) that fills the whole window, with white glass panels floating over it. On the left: what is happening, the market panel, the districts, the Adaptive Quarter and this step. On the right: the selected unit with its values, its decision and the town log. Along the top: capital, sentiment, volume and price, day and step, the live lamp and the links to the other views. On the map: the market bar, the ticker, a command bar with Shock (Calm / Boom / Crash) and Speed (pause, step, 1× / 4× / 16×), a key and the camera buttons. Along the bottom: Scenario, Experiment, Headline, Controls, Replay and Restart, the next strategy review and the scale switch EURO / COMPARE.

On the map	Means
Block height of a district	the district's capital, on the euro ladder; the footer states the scale (“1 floor = €1M” and so on)
Road width	order flow this step
Chevrons moving along a road	the direction of that flow, towards the exchanges
A walking figure	ten agents carrying an order
The crate it carries	green (buy) or red (sell), in the proportion of that group's buy orders this step

On the map	Means
Golden figures between the two camps	agents switching strategy
Trucks	the ETF Depot's scheduled purchases and the Issuer's new shares
Smoke from the Issuer's chimneys	it sold stock this step
Five blue towers	the five exchanges. Height = book depth; a band shows symbol and price; the selected market's tower is brighter
Numbers rising over the selected tower	its fills
Hatched upper floors of the Leverage Tower	borrowed capital
Red beacon blinking fast, red ring pulsing on the ground, MARGIN CALL tag	a margin call
Blue pin	the selected district
Light and weather	sentiment: warm sun when greedy, overcast when fearful, rain below about 30
Trees, the faint town on the horizon	nothing

- **Operating the town**: click a building or its label to select the district; a tower selects that market. Drag to turn the camera, scroll to zoom, right-drag (or shift-drag) to pan. **+** – **0** zoom and reset, and a button turns the town by 45°. Without WebGL the page says so, and the panels keep working.
- **Controllable from the town**: load a **scenario** (including the five historical episodes), start an **experiment** and read its measured check, inject a **headline** by hand, move any **control**, open a **recording**, restart the run. The `?` at the top left says what the town shows and what to try first. A click on a district explains it and puts a real member into the unit card.
- **Replay in the town**: a recorded run is recomputed from its seed, not read from curves. That is why the whole state is available at every position, and the slider moves the complete map including the order book.
- **Two scales**: after a hundred trading days one camp holds almost all the capital, the spread reaches 135×, and on the euro scale the other districts all sit at minimum height. **EURO** stays the honest default; **COMPARE** is logarithmic and says so in the same line.
- **Price, candles, order book**: the market panel on the left shows the same instrument on three levels: price against fair value, daily candles and the resting orders. The book is where a price is made in this model; almost all of the depth sits at the touch, where the market makers put it.
- **The analysis page** (`/sectors`): sector tiles with sparklines, an instrument panel with candles, order book and tape, correlation heatmap, Central Bank with duration per sector, validation panel with variance ratio, A/B comparison, replay slider.
- **The index board** (`/indices`): all indices live on one axis, normalised to the start of the window, and for each index its level, fair-value index, volatility, drawdown, breadth and the *measured* beta against the one set in the factor model. An index here is a measurement: weighted by the shares the ETF itself buys, and chain-linked.
- **The trader list** (`/traders`): all agents side by side instead of one after another: capital, cash, positions, leverage, result since the start of the run, forced sales. Sorted and filtered in the backend, with rollups per district and a jump back into the town to exactly this agent.
- **Every parameter** is declared once (`PARAM_SPEC`) and the control panel is generated from it. A new parameter appears without any change to the frontend.

Measure and compare

Tool	Answers
`calibrate.py`	Does the model meet the six criteria, per instrument, across several seeds?
`sweep.py ablate`	Which mechanism is responsible for which property? Switches each one off in turn.
`sweep.py param`	How does a measure depend on a parameter, across a range of values?
`/api/compare`	What does this parameter do, from the same random start?
`/api/compare/sweep`	Is the difference larger than the scatter between seeds? Returns signal/noise.
`run.py`	A run without the interface, exported as Parquet for analysis in a notebook.
`/api/replay`	Scrub freely through a recorded run.
`/api/explain`	What is happening, why, what to watch, as text.
`/api/lessons/{k}/check`	Did the experiment show what it set out to show? Measured.

Why the A/B comparison counts now, and did not before

Every agent draws from **its own** random stream, seeded from the seed and the agent ID. With a shared stream, the number of random draws consumed depended on what an agent decided. The first difference in behaviour threw everyone out of step, and a channel worth 0.4% became a 39% difference. Now the effect scales with its cause: a borrowing spread of +5 basis points gives a 0.4% gap, trend following from 55 to 140 gives 3.5%. And a parameter that *cannot* have an effect (a leverage limit without leveraged funds) produces bit-identical runs. This control test is in the suite.

Scenarios

A scenario never writes a price path. It sets conditions and supplies headlines. What the price then does is the model's answer.

- **Basic scenarios:** Calm market · Bubble formation · Leverage crash · ETF dominance · News shock
- **Historical episodes:** Dotcom bubble 1998–2000 · Financial crisis 2008 · COVID crash 2020 · Meme stock 2021 · Rate turn 2022

The historical episodes reproduce **mechanisms**, not price paths. Turning parameters until a chart matches history would prove nothing. The question is narrower and can be answered: does the mechanism credited with this episode produce that kind of behaviour in a model that was not built for it?

7 · The method, and why it makes the difference

The most instructive part of this project is not the result but the way there. Almost every plausible explanation was wrong.

The test that explained everything

The **variance ratio**, Var(k-period return) / (k · Var(1-period)), separates permanent from transitory variance. A random walk gives 1 at every horizon.

```
20 steps 5.79 strongly trending
240 steps 0.72 one day, already reverting
4800 steps 0.066 twenty days
```

0.066 means the variance of 20-day returns was 6.6% of what a random walk would give. The price was nailed to fair value: 79% price volatility against 15% fundamental volatility, so around 90% of all price movement was transitory. And transitory movement *has to* be taken back. That is where the negative autocorrelation came from. No single agent type was to blame.

What was ruled out

Hypothesis	Test	Result
Market makers shift prices with their inventory	remove them completely	ACF stays at -0.40
Constant rebalancing (target = wealth ÷ price)	tolerance band 5–30%	no effect, other criteria worse
Issuer overcorrects (bang-bang control)	proportional issuance	no effect
Value investors correct too fast	pull-back strength 0.08 instead of 0.005	ACF gets *worse* (-0.22)
Trading day too short	10 / 30 / 120 / 360 steps per day	ACF(1) around -0.4 throughout
Measurement noise in the closing price	averaging over the last 5% of the day	-0.43 becomes -0.39
Valuation is recognised too quickly	delay of 0.5 / 12 / 30 days	**worse**: the price trends instead of reverting, correlation falls from 0.42 to 0.13, and the cascade stops firing

Two of these conclusions replaced earlier ones drawn from the same experiments *before* a fix to strategy switching. The dynamics underneath had changed, which is a good argument for re-running an ablation rather than trusting a note. That is what `scripts/sweep.py ablate` is for.

What worked

- **Volatility clustering and mean reversion were the same mechanism.** The clustering came from regime switches, and the switch back to value *is* the reversion. Every parameter that improved the autocorrelation destroyed the clustering. The solution was a second, independent source of clustering: news arriving in bursts (the volatility of fair value is itself persistent).
- **Activity follows the news flow** (Clark 1973): volume and volatility are both consequences of the rate of information, not consequences of each other.
- **A market's capital has to scale with the number of markets.** With five sectors each market saw a fifth of the capital, while fair value diffused just as fast. Prices could not keep up, and the correlation from the factor model barely reached them: 0.07 instead of a realistic 0.4–0.6. After scaling: **0.40**.
- **Margin calls were dead code under the defaults.** The funds reach ~1.6× leverage; a threshold of 0.25 is never touched. It is now set against the leverage the model actually produces.

8 · Where the model is wrong

In full, because a simulation cannot be judged without this list.

Finding	How far from the target	Diagnosis
Returns are predictable	ACF -0.22 instead of ±0.1	The best value reached was -0.03 on one seed, in a configuration with a sector correlation of 0.19 and no margin calls. The trade-off is not solved.
Volume does not follow volatility	-0.07 instead of > 0.1	Most volume is market-maker volume and does not rise with agitation. Countermeasures made four other measures worse.
Volatility too high	1.31 a year against 0.15–0.25 in reality	Partly a consequence of the active cascade: forced selling creates volatility.
One sector far from fair value	Financials 89%, the others below 40%	The Issuer only steps in above its threshold; in a long-running sector the flow is not enough to keep up.
Kurtosis unstable	5.4 ± 5.9	Driven by rare, very large single days in thin sectors, not by consistently heavy tails.

Not built

- **No dividends, no earnings.** Fair value is an exogenous process, not something a company earns. Buybacks are paid from a cash pile that never fills from the business.
- **The four views stay separate.** Town, analysis, indices and traders show the same simulation from four angles. Nothing was merged, because the town is readable precisely because it draws one market.
- **No reinforcement learning and no LLM agents.** Dropped on purpose: both cost reproducibility, and without a fixed seed no run is good enough as a result.

9 · Command reference

```
./start.sh start the interface

python scripts/calibrate.py --runs 5 --steps 90000 validation, per instrument
python scripts/sweep.py ablate --steps 80000 switch off each mechanism in turn
python scripts/sweep.py param --name n_passive \
--values 0,50,100,200 sweep one parameter
python scripts/run.py --steps 120000 --name baseline run without UI, Parquet export
python scripts/run.py --scenario gfc2008 --name crisis

python -m pytest tests/ -q 324 tests
```

Where things are in the code

File	Contents
`mes/orderbook.py`	matching engine, price-time priority: the foundation
`mes/agents.py`	the seven decision rules
`mes/simulation.py`	the loop, settlement, margin, strategy switching
`mes/market.py`	one instrument: book, history, daily bars
`mes/instruments.py`	the five sectors and the factor model

File	Contents
`mes/centralbank.py`	Taylor rule, macro block, the three channels
`mes/metrics.py`	the stylized-facts checks and the variance ratio
`mes/compare.py`	A/B with a signal/noise assessment
`mes/narrator.py`	state becomes explanation: what, why, what to watch
`mes/lessons.py`	the seven experiments, each with a measured check
`mes/config.py`	every parameter, declared once
`web/index.html`	the town: panels, and the model it builds from /api/state (no build step)
`web/town3d.js`	the 3D renderer; draws only the model index.html builds
`web/vendor/`	three.js r159 (MIT), vendored as a classic script so the page works offline and from file://
`web/sectors.html`	the analysis page
`web/indices.html`	the index board: all indices live against each other
`web/traders.html`	the trader list: all agents, sortable and filterable
`mes/indices.py`	the index construction: chain-linked, ETF-weighted
`mes/traders.py`	population to table: the only translation point

Literature

Cont (2001), *Empirical properties of asset returns* · Lux & Marchesi (1999), *Scaling and criticality in a stochastic multi-agent model* · Kirman (1993), *Ants, rationality, and recruitment* · Brock & Hommes (1997), *A rational route to randomness* · Chiarella & Iori (2002), *A simulation analysis of the microstructure of double auction markets* · Clark (1973), *A subordinated stochastic process model …* · Lo & MacKinlay (1988), *Stock market prices do not follow random walks* · Taylor (1993), *Discretion versus policy rules in practice* · LeBaron (2006), *Agent-based computational finance*